

5.1-Eigenvalues & Eigenvectors

$$\begin{array}{ccccccc} A & X & = & \lambda & X \\ \downarrow & \downarrow & & \downarrow & \downarrow \\ \text{Matrix} & \text{vectors} & & \text{scalar} & \text{vector} \\ & & & \text{(Eigen value)} & \end{array}$$

→ Eigenvector is non-zero vector.

→ Eigen value can't be 0.

$$AX = \lambda X$$

$$\lambda X - AX = 0$$

$$(\lambda I - A) = 0$$

Example: 1 (Eigen vector of 2×2 Matrix)

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \text{ is eigenvector}$$

$$A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$$

$$Ax = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

$$Ax = 3 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 3x$$

Example: 2 (Finding Eigen Values)

$$A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$$

$$\det(\lambda I - A) = 0$$

$$\lambda I - A =$$

$$\lambda I - A = \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$$

$$\lambda I - A = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$$

$$\lambda I - A = \begin{bmatrix} \lambda - 3 & 0 \\ -8 & \lambda - 1 \end{bmatrix}$$

$$\det(\lambda I - A) = \det \begin{bmatrix} \lambda - 3 & 0 \\ -8 & \lambda - 1 \end{bmatrix}$$

$$\det(\lambda I - A) = (\lambda - 3)(\lambda - 1) - 0$$

$$\det(\lambda I - A) = (\lambda - 3)(\lambda - 1)$$

$$\det(\lambda I - A) = 0$$

$$(\lambda - 3)(\lambda - 1) = 0$$

$$\lambda - 3 = 0, \lambda - 1$$

$$\boxed{\lambda = 3}, \boxed{\lambda = 1}$$

Example: 3

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{bmatrix}$$

$$\lambda I - A :-$$

$$= \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} \lambda - 1 & 0 & 0 \\ 0 & \lambda - 1 & 0 \\ -4 & 17 & \lambda - 8 \end{bmatrix}$$

$$\det(\lambda I - A) :-$$

$$= \det \begin{vmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ -4 & 17 & \lambda - 8 \end{vmatrix}$$

$$= \lambda(\lambda^2 - 8\lambda + 17) + 1(0 - 4) + 0$$

$$= \lambda^3 - 8\lambda^2 + 17\lambda - 4$$

$$\det(\lambda I - A) = 0$$

$$\lambda^3 - 8\lambda^2 + 17\lambda - 4 = 0$$

↳ We need to find such value that

use $\lambda = 4$ is factor or multiple of 4

$$(4)^3 - 8(4)^2 + 17(4) - 4 = 0 \text{ and make LHS } 0$$

$$64 - 128 + 68 - 4 = 0$$

$$0 = 0$$

so $(\lambda - 4)$ is a factor

use Synthetic Division:

$$\lambda^3 - 8\lambda^2 + 17\lambda - 4 = 0$$

4	1	-8	17	-4
	↓	4	-16	4
	1	-4	1	0

$$\lambda = 4$$

$$\lambda^2 - 4\lambda + 1 = 0$$

$$(\lambda - 4) = 0$$

$$(\lambda - 4)(\lambda^2 - 4\lambda + 1) = 0$$

$$\lambda - 4 = 0$$

$$\lambda = 4$$

$$\lambda^2 - 4\lambda + 1 = 0$$

$$a=1, b=-4, c=1$$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \Rightarrow 4 \pm \frac{\sqrt{16-4}}{2}$$

$$\lambda = \frac{4 \pm \sqrt{12}}{2} \Rightarrow \frac{4 \pm 2\sqrt{3}}{2}$$

$$\lambda = 2(2 \pm \sqrt{3})$$

$$\lambda = 2 \pm \sqrt{3}$$

$$\boxed{\lambda = 4}, \boxed{\lambda = 2 + \sqrt{3}}, \boxed{\lambda = 2 - \sqrt{3}}$$

Example: 4 (Eigenvalues of an upper triangular matrix)

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{bmatrix}$$

$$\lambda I - A =$$

$$= \lambda \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{bmatrix}$$

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$$= \begin{bmatrix} \lambda - a_{11} & -a_{12} & -a_{13} & -a_{14} \\ 0 & \lambda - a_{22} & -a_{23} & -a_{24} \\ 0 & 0 & \lambda - a_{33} & -a_{34} \\ 0 & 0 & 0 & \lambda - a_{44} \end{bmatrix}$$

$\det(\lambda I - A) :-$

$$= \det \begin{bmatrix} \lambda - a_{11} & -a_{12} & -a_{13} & -a_{14} \\ 0 & \lambda - a_{22} & -a_{23} & -a_{24} \\ 0 & 0 & \lambda - a_{33} & -a_{34} \\ 0 & 0 & 0 & \lambda - a_{44} \end{bmatrix}$$

$\rightarrow$ AS it's triangular matrix.

$$= (\lambda - a_{11})(\lambda - a_{22})(\lambda - a_{33})(\lambda - a_{44})$$

$$\det(\lambda I - A) = 0$$

$$(\lambda - a_{11})(\lambda - a_{22})(\lambda - a_{33})(\lambda - a_{44}) = 0$$

$$\lambda = a_{11}, \lambda = a_{22}, \lambda = a_{33}, \lambda = a_{44}$$

Example: 5 (Eigenvalues of Lower Triangular matrix)

$$A = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ -1 & \frac{2}{3} & 0 \\ 5 & -8 & -\frac{1}{4} \end{bmatrix}$$

$$\lambda = \frac{1}{2}, \lambda = \frac{2}{3}, \lambda = -\frac{1}{4}$$

So this example shows if a matrix is triangular (either lower or upper), we don't need to do any complex calculations to find its eigen values - they are simply the entries sitting on main diagonal.

Example: 6- (Bases For Eigenspaces or Find eigen vectors)

Step 1: Find eigen value.

$$A = \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix}$$

$$\lambda I - A =$$

$$= \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \lambda + 1 & -3 \\ -2 & \lambda \end{bmatrix} \rightarrow \textcircled{1}$$

$$\det(\lambda I - A) = 0$$

$$\det \begin{bmatrix} \lambda + 1 & -3 \\ -2 & \lambda \end{bmatrix} = 0$$

$$\lambda(\lambda + 1) - 6 = 0$$

$$\lambda^2 + \lambda - 6 = 0$$

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$$\lambda^2 + 3\lambda - 2\lambda - 6 = 0$$

$$\lambda(\lambda + 3) - 2(\lambda + 3) = 0$$

$$(\lambda - 2)(\lambda + 3) = 0$$

$$\boxed{\lambda = 2}, \quad \boxed{\lambda = -3}$$

Step: 2 - Find Eigen vectors.

$$(\lambda I - A) X = 0$$

use gr (1)

$$\begin{bmatrix} \lambda + 1 & -3 \\ -2 & \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

For $\lambda = 2$

$$\begin{bmatrix} 3 & -3 & 0 \\ -2 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 0 \\ -2 & 2 & 0 \end{bmatrix} \begin{matrix} R_1 \\ 2 \end{matrix}$$

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad R_2 + 2R_1$$

Leading Var = x_1

Let

Free Var = x_2

$$x_2 = t$$

So

$$x_1 - x_2 = 0$$

$$x_1 = t$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} t \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$E_1 = \left\langle \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\rangle$$

For $\lambda = -3$

$$\begin{bmatrix} -2 & -3 & 0 \\ -2 & -3 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3/2 & 0 \\ -2 & -3 & 0 \end{bmatrix} \begin{matrix} R_1 \\ -R_1 \end{matrix}$$

$$\begin{bmatrix} 1 & 3/2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad R_2 + 2R_1$$

Leading Var = x_1 Let $x_2 = t$

Free Var = x_2

$$x_1 + 3/2 x_2 = 0$$

$$x_1 = -3/2 t$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -3/2 t \\ t \end{bmatrix} = t \begin{bmatrix} -3/2 \\ 1 \end{bmatrix}$$

$$E_2 = \left\langle \begin{bmatrix} -3/2 \\ 1 \end{bmatrix} \right\rangle$$

Some Notes:

1. In characteristic equation last constant term will be determinant.
2. Product of eigenvalues will produce determinant.

Exercise Set 5-1

Question: 1-4

Confirm by multiplication that x is an eigenvector of A , and find corresponding eigenvalue.

$$AX = \lambda x$$

2-

$$A = \begin{bmatrix} 5 & -1 \\ 1 & 3 \end{bmatrix}, \quad x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$AX = \begin{bmatrix} 5 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5-1 \\ 1+3 \end{bmatrix}$$

$$AX = \begin{bmatrix} 4 \\ 4 \end{bmatrix} \Rightarrow AX = 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$AX = 4X$$

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Since $AX = 4X$, x is eigenvector and corresponding eigenvalue is $\lambda = 4$

4-

$$A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}, x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$AX = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$AX = \begin{bmatrix} 2-1-1 \\ -1+2-1 \\ -1-1+2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$AX = 0 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$AX = 0X$$

Since $AX = 0X$, x is eigenvector (note eigenvector can't be 0 themselves an eigenvalue can be 0) and corresponding eigenvalue is $\lambda = 0$

1-3

will be solved in same way.

Example : 7

Find eigen vectors / eigen space of

$$A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ -1 & 0 & 3 \end{bmatrix}$$

1. Eigenvalues:-

$$\lambda I - A :-$$

$$= \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} \lambda & 0 & 2 \\ -1 & \lambda-2 & -1 \\ -1 & 0 & \lambda-3 \end{bmatrix} \rightarrow (1)$$

$$\det(\lambda I - A) = 0$$

$$\det \begin{bmatrix} \lambda & 0 & 2 \\ -1 & \lambda-2 & -1 \\ -1 & 0 & \lambda-3 \end{bmatrix} = 0$$

$$\lambda [(\lambda-2)(\lambda-3)+0] - 0 + 2 [0 - (-1)(\lambda-2)] = 0$$

$$\lambda(\lambda-2)(\lambda-3) + 2(\lambda-2) = 0$$

$$(\lambda-2)[\lambda(\lambda-3)+2] = 0$$

$$\lambda-2=0, \quad \lambda(\lambda-3)+2=0$$

$$\lambda=2, \quad \lambda^2-3\lambda+2=0$$

$$\lambda^2-2\lambda-\lambda+2=0$$

$$\lambda(\lambda-2)-1(\lambda-2)=0$$

$$(\lambda-2)(\lambda-1)=0$$

$$\lambda-2=0, \quad \lambda-1=0$$

$$\lambda=2, \quad \lambda=1$$

2. Eigen vectors:-

For $\lambda=2$

$$(\lambda I - A)X = 0$$

From eq (2)

$$\begin{bmatrix} 2 & 0 & 2 \\ -1 & 0 & -1 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 & 2 & 0 \\ -1 & 0 & -1 & 0 \\ -1 & 0 & -1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 0 & -1 & 0 \\ -1 & 0 & -1 & 0 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

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$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{l} R_2+R_1 \\ R_3+R_1 \end{array}$$

Leading var = x_1 Let

Free var = x_2, x_3 $x_2 = t, x_3 = s$

$$x_1 + x_3 = 0$$

$$x_1 = -s$$

$$x = \begin{bmatrix} -s \\ t \\ s \end{bmatrix} = \begin{bmatrix} -s \\ 0 \\ s \end{bmatrix} + \begin{bmatrix} 0 \\ t \\ 0 \end{bmatrix}$$

$$x = s \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$E_1 = \left\langle \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\rangle$$

For $\lambda = 1$

$$\begin{bmatrix} 1 & 0 & 2 \\ -1 & -1 & -1 \\ -1 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ -1 & -1 & -1 & 0 \\ -1 & 0 & -2 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{l} R_2+R_1 \\ R_3+R_1 \end{array}$$

Leading var = x_1, x_2 Let

Free var = x_3 $x_3 = t$

$$x_1 + 2x_2 = 0$$

$$-x_2 + x_3 = 0$$

$$-x_2 + x_3 = 0$$

$$x_1 + 2x_2 = 0$$

$$-x_2 + t = 0$$

$$x_1 + 2t = 0$$

$$t = x_2$$

$$x_1 = -2t$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2t \\ t \\ t \end{bmatrix} = t \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

$$E_2 = \left\langle \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\rangle$$

Example: 8 (Eigenvalues & Invertibility)

If eigenvalue becomes 0 then matrix is not invertible and if non zero then it invertible. verify using example 7.

Question : 5-6.

Find Characteristic equation, eigen values and bases for eigenspaces of matrix.

Question : 5

b-

$$\begin{bmatrix} -2 & -7 \\ 1 & 2 \end{bmatrix}$$

$$\det(\lambda I - A) = 0$$

1. Eigenvalues

$$\lambda I - A :-$$

$$= \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -2 & -7 \\ 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} \lambda + 2 & 7 \\ -1 & \lambda - 2 \end{bmatrix} \rightarrow (1)$$

$$\det(\lambda I - A) = 0$$

$$\det \begin{bmatrix} \lambda + 2 & 7 \\ -1 & \lambda - 2 \end{bmatrix} = 0$$

$$(\lambda + 2)(\lambda - 2) + 7 = 0$$

$$\lambda^2 - 2\lambda + 2\lambda - 4 + 7 = 0$$

$$\lambda^2 + 3 = 0$$

$$\lambda^2 = -3$$

$$\lambda = \pm \sqrt{-3}$$

$$\lambda_1 = +\sqrt{3}i, \lambda_2 = -\sqrt{3}i$$

$\rightarrow$ characteristic eq.

2. Eigenvectors

$$(\lambda I - A)x = 0$$

using eq (1)

$$\begin{bmatrix} \lambda + 2 & 7 \\ -1 & \lambda - 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

∴

For $\lambda = i\sqrt{3}$

$$= \begin{bmatrix} i\sqrt{3} + 2 & 7 & 0 \\ -1 & i\sqrt{3} - 2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & i\sqrt{3} - 2 & 0 \\ i\sqrt{3} + 2 & 7 & 0 \end{bmatrix} \quad R_1 \leftrightarrow R_2$$

$$= \begin{bmatrix} 1 & 2 - i\sqrt{3} & 0 \\ i\sqrt{3} + 2 & 7 & 0 \end{bmatrix} \quad -R_1$$

$$= \begin{bmatrix} 1 & 2 - i\sqrt{3} & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad R_2 - (i\sqrt{3} + 2)R_1$$

leading var = x_1

Free var = x_2

let $x_2 = t$

$$x_1 + 2 - i\sqrt{3}x_2 = 0$$

$$x_1 + 2 - i\sqrt{3}t = 0$$

$$7 - (2 + i\sqrt{3})(2 - i\sqrt{3})$$

$$7 - (2^2 - (\sqrt{3})^2 i^2)$$

$$7 - (4 - 3(-1))$$

$$7 - (4 + 3)$$

$$7 - 7 = 0$$

$$x_2 = -2 + \sqrt{3}i; t$$

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2 + \sqrt{3}i; t \\ t \end{bmatrix} = t \begin{bmatrix} -2 + \sqrt{3}i \\ 1 \end{bmatrix}$$

$$E_1 = \left\langle \begin{bmatrix} -2 + \sqrt{3}i \\ 1 \end{bmatrix} \right\rangle$$

For $\lambda = -\sqrt{3}i$

$$= \left[\begin{array}{cc|c} -\sqrt{3}i + 2 & 1 & 0 \\ -1 & -\sqrt{3}i - 2 & 0 \end{array} \right]$$

$$= \left[\begin{array}{cc|c} -1 & -2 - \sqrt{3}i & 0 \\ 2 - \sqrt{3}i & 1 & 0 \end{array} \right]$$

$$R_1 \leftrightarrow R_2$$

$$= \left[\begin{array}{cc|c} 1 & 2 + \sqrt{3}i & 0 \\ 2 - \sqrt{3}i & 1 & 0 \end{array} \right] -R_1$$

$$= \left[\begin{array}{cc|c} 1 & 2 + \sqrt{3}i & 0 \\ 0 & 0 & 0 \end{array} \right] R_2 = (2 - \sqrt{3}i)R_1$$

Free var = x_2 Let $x_2 = t$

Lead var = x_1

$$x_1 + 2 + \sqrt{3}ix_2 = 0$$

$$x_1 + 2 + \sqrt{3}it = 0$$

$$x_1 = -2 - \sqrt{3}it = 0$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2-\sqrt{3}it \\ t \end{bmatrix} = t \begin{bmatrix} -2-\sqrt{3}i \\ 1 \end{bmatrix}$$

$$E_2 = \left\langle \begin{bmatrix} -2-\sqrt{3}i \\ 1 \end{bmatrix} \right\rangle$$

a, c, d-

solved in same way

Question: 6-

C-

$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

1. Eigenvalues

$$\det(\lambda I - A) = 0$$

$$\lambda I - A =$$

$$= \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} \lambda - 2 & 0 \\ 0 & \lambda - 2 \end{bmatrix}$$

$$\det (\lambda I - A) = 0$$

$$-b, d=0$$

$$\det \begin{bmatrix} \lambda - 2 & 0 \\ 0 & \lambda - 2 \end{bmatrix} = 0$$

$$(\lambda - 2)(\lambda - 2) - 0 = 0 \text{ or } (\lambda - 2)^2 = 0 \rightarrow \text{Charac Equation.}$$

$$\lambda - 2 = 0, \lambda - 2 = 0 \quad \lambda^2 - 2\lambda + 4 = 0$$

$$\lambda = 2, \lambda = 2$$

2. Eigenvectors

$$(\lambda I - A)x = 0$$

$$\begin{bmatrix} \lambda - 2 & 0 \\ 0 & \lambda - 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

For $\lambda = 2$

$$\left[\begin{array}{cc|c} 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

Free var = x_1, x_2

let $x_1 = t$
 $x_2 = s$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} t \\ s \end{bmatrix} = \begin{bmatrix} t \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ s \end{bmatrix}$$

$$x = t \begin{bmatrix} 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$E_1 = \left\langle \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\rangle$$

a-b, d-

same as part c.

Question: 7-12

Find characteristic equation, Eigen values and eigenvectors.

7-

$$\begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

1. Eigen Values:-

$$\det(\lambda I - A) = 0$$

$$\lambda I - A:-$$

$$= \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \lambda - 4 & 0 & -1 \\ 2 & \lambda - 1 & 0 \\ 2 & 0 & \lambda - 1 \end{bmatrix}$$

$$\det(\lambda I - A) = 0$$

$$\det \begin{bmatrix} \lambda - 4 & 0 & -1 \\ 2 & \lambda - 1 & 0 \\ 2 & 0 & \lambda - 1 \end{bmatrix} = 0$$

expand along C_2

$$0 + (-1)^{2+2} \begin{vmatrix} \lambda-4 & -1 \\ 2 & \lambda-1 \end{vmatrix} + 0 = 0$$

$$(\lambda-1) [(\lambda-1)(\lambda-4)+2] = 0 \quad (1)$$

To Find characteristic Equation:-

$$(\lambda-1) [\lambda^2 - 4\lambda - \lambda + 4 + 2] = 0$$

$$(\lambda-1) (\lambda^2 - 5\lambda + 6) = 0$$

$$(\lambda^3 - 5\lambda^2 + 6\lambda - \lambda^2 + 5\lambda - 6) = 0$$

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

↳ characteristic eq.

Use eq. 1

$$\lambda - 1 = 0$$

$$\lambda = 1$$

$$(\lambda-1)(\lambda-4)+2=0$$

$$\lambda^2 - 4\lambda - \lambda + 4 + 2 = 0$$

$$\lambda^2 - 5\lambda + 6 = 0$$

$$\lambda^2 - 3\lambda - 2\lambda + 6 = 0$$

$$\lambda(\lambda-3) - 2(\lambda-3) = 0$$

$$(\lambda-2)(\lambda-3) = 0$$

$$(\lambda-2)=0, \lambda-3=0$$

$$\lambda=2, \lambda=3$$

2. Eigen vectors:

$$(\lambda I - A) X = 0$$

$$\begin{bmatrix} \lambda-4 & 0 & -1 \\ 2 & \lambda-1 & 0 \\ 2 & 0 & \lambda-1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

For $\lambda = 1$

$$= \left[\begin{array}{ccc|c} -3 & 0 & -1 & 0 \\ 2 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \end{array} \right]$$

$$= \left[\begin{array}{ccc|c} 2 & 0 & 0 & 0 \\ -3 & 0 & -1 & 0 \\ 2 & 0 & 0 & 0 \end{array} \right] \quad R_1 \leftrightarrow R_2$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ -3 & 0 & -1 & 0 \\ 2 & 0 & 0 & 0 \end{array} \right] \quad \begin{array}{l} R_1 \\ \hline 2 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \begin{array}{l} R_2 + 3R_1 \\ R_3 - 2R_1 \end{array}$$

Leading var = x_1, x_3 Let
Free var = x_2 $x_2 = t$

$$x_1 = 0$$

$$-x_3 = 0 \Rightarrow x_3 = 0$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ t \\ 0 \end{bmatrix} = t \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$E_1 = \left\langle \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\rangle$$

For $\lambda = 2$:

$$= \left[\begin{array}{ccc|c} -2 & 0 & -1 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{2} & 0 \\ 2 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 \end{array} \right] \begin{array}{l} R_1 \\ R_2 - 2R_1 \\ R_3 - 2R_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{l} R_2 - 2R_1 \\ R_3 - 2R_1 \\ \end{array}$$

Leading Var = x_1, x_2 Let

Free Var = x_3 $x_3 = t$

$$\begin{array}{l|l} x_1 + \frac{1}{2}x_3 = 0 & x_2 - x_3 = 0 \\ x_1 = -\frac{1}{2}x_3 & x_2 = x_3 \\ x_1 = -\frac{1}{2}t & x_2 = t \end{array}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}t \\ t \\ t \end{bmatrix} = t \begin{bmatrix} -\frac{1}{2} \\ 1 \\ 1 \end{bmatrix}$$

$$E_2 = \left\langle \begin{bmatrix} -\frac{1}{2} \\ 1 \\ 1 \end{bmatrix} \right\rangle$$

For $\alpha = 3$

$$= \left[\begin{array}{ccc|c} -1 & 0 & -1 & 0 \\ 2 & 2 & 0 & 0 \\ 2 & 0 & 2 & 0 \end{array} \right]$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 2 & 2 & 0 & 0 \\ -2 & 0 & 2 & 0 \end{array} \right] \begin{array}{l} -R_1 \\ \\ \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{l} R_2 - 2R_1 \\ R_3 - 2R_1 \\ \end{array}$$

Leading var = x_1, x_2 LetFree var = x_3 $x_3 = t$

$$x_1 + x_3 = 0$$

$$x_1 = -x_3$$

$$x_1 = -t$$

$$2x_2 - 2x_3 = 0$$

$$2x_2 = 2x_3$$

$$x_2 = t$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -t \\ t \\ t \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$$E_3 = \left\langle \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\rangle$$

12-

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$$

1. Eigenvalues

$$\det(\lambda I - A) = 0$$

$$\lambda I - A = ?$$

$$= \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} \lambda - 1 & 3 & -3 \\ -3 & \lambda + 5 & -3 \\ -6 & 6 & \lambda - 4 \end{bmatrix}$$

$$\det(\lambda I - A) = 0$$

$$\det \begin{bmatrix} \lambda - 1 & 3 & -3 \\ -3 & \lambda + 5 & -3 \\ -6 & 6 & \lambda - 4 \end{bmatrix} = 0$$

Expand along Row 1

$$\lambda - 1 \begin{vmatrix} \lambda + 5 & -3 \\ 6 & \lambda - 4 \end{vmatrix} - 3 \begin{vmatrix} -3 & -3 \\ -6 & \lambda - 4 \end{vmatrix} - 3 \begin{vmatrix} -3 & \lambda + 5 \\ -6 & 6 \end{vmatrix} = 0$$

$$(\lambda - 1) [(\lambda + 5)(\lambda - 4) + 18] - 3 [-3(\lambda - 4) - 18] - 3 [-18 + 6(\lambda + 5)] = 0$$

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$$(x-1)[x^2 - 4x + 5x - 20 + 18] - 3[-3x + 12 + 18]$$

$$-3[-18 + 6x + 30] = 0$$

$$(x-1)(x^2 + x - 2) - 3(-3x - 6) - 3(6x + 12) = 0$$

$$x^3 + x^2 - 2x - x^2 - x + 2 + 9x + 18 - 18x - 36 = 0$$

$$x^3 - 12x - 16 = 0$$

↳ Characteristic Eq

Use multiple of 16 to make LHS = 0

use $x = -2$

$$x^3 - 12x - 16 = 0$$

$$(-2)^3 - 12(-2) - 16 = 0$$

$$-8 + 24 - 16 = 0$$

$$-24 + 24 = 0$$

$$0 = 0$$

so $(x+2)$ is our factor

Use Synthetic Division

$$x^3 - 12x - 16$$

-2	1	0	-12	-16
	↓	-2	4	16
	1	-2	-8	0

$$x^2 - 2x - 8 = 0$$

$$x^2 - 4x + 2x - 8 = 0$$

$$x(x-4) + 2(x-4) = 0$$

$$(\lambda + 2)(\lambda - 4) = 0$$

$$\lambda + 2 = 0, \quad \lambda - 4 = 0$$

$$\lambda = -2, \quad \lambda = 4$$

So Eigenvalues are

$$\lambda = -2, \quad \lambda = -2, \quad \lambda = 4$$

2. Eigenvectors.

$$(\lambda I - A) = 0$$

$$\begin{bmatrix} \lambda - 1 & 3 & -3 \\ -3 & \lambda + 5 & -3 \\ -6 & 6 & \lambda - 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

For $\lambda = -2$

$$= \begin{bmatrix} -3 & 3 & -3 & | & 0 \\ -3 & 3 & -3 & | & 0 \\ -6 & 6 & -6 & | & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -1 & 1 & | & 0 \\ -3 & 3 & -3 & | & 0 \\ -6 & 6 & -6 & | & 0 \end{bmatrix} \quad \begin{matrix} \\ -R_2 \\ -R_3 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & -1 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \quad \begin{matrix} R_2 + 3R_1 \\ R_3 + 6R_1 \end{matrix}$$

Leading var = x_1

Let

Free var = x_2, x_3

$x_2 = t, x_3 = s$

$$x_1 - x_2 + x_3 = 0$$

$$x_1 - t + s = 0$$

$$x_1 = t - s$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} t-s \\ t \\ s \end{bmatrix} = \begin{bmatrix} t \\ t \\ 0 \end{bmatrix} + \begin{bmatrix} -s \\ 0 \\ s \end{bmatrix}$$

$$X = t \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$E_1 = \left\langle \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\rangle$$

For $n = 4$

$$= \begin{bmatrix} 3 & 3 & -3 & 0 \\ -3 & 9 & -3 & 0 \\ -6 & 6 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & -1 & 0 \\ -3 & 9 & -3 & 0 \\ -6 & 6 & 0 & 0 \end{bmatrix} \begin{matrix} -R_1 \\ 3 \\ 3 \end{matrix}$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 12 & -6 & 0 \\ 0 & 12 & -6 & 0 \end{array} \right] \begin{array}{l} R_2 + 3R_1 \\ R_3 + 6R_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 12 & -6 & 0 \end{array} \right] \begin{array}{l} R_2 \times \frac{1}{12} \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{l} R_3 - 12R_2 \end{array}$$

Leading var = x_1, x_2 Let

Free var = x_3 $x_3 = t$

$$x_1 + x_2 - x_3 = 0$$

$$x_1 = -x_2 + x_3$$

$$x_1 = -\frac{1}{2}t + t$$

$$x_1 = \frac{1t + 2t}{2}$$

$$x_1 = \frac{1}{2}t$$

$$x_2 - \frac{1}{2}x_3 = 0$$

$$x_2 = \frac{1}{2}x_3$$

$$x_2 = \frac{1}{2}t$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}t \\ \frac{1}{2}t \\ t \end{bmatrix} = t \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 1 \end{bmatrix}$$

$$E_1 = \left\langle \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 1 \end{bmatrix} \right\rangle$$

8-11

Same as Q7 and Q12.

Question : 24

Find $\det(A)$ given that A has $p(\lambda)$ as its characteristic polynomial.

(Hint: See Theorem 5.1.4)

Making Formula

Characteristic Polynomial of $n \times n$ matrix is defined as

$$p(\lambda) = \det(\lambda I - A)$$

If we put $\lambda = 0$ in equation

$$p(0) = \det(0I - A)$$

$$p(0) = \det(-A)$$

$$p(0) = (-1)^n \det(A) \quad \therefore \det(kA) = k^n \det(A)$$

$$\det(A) = \frac{p(0)}{(-1)^n} \rightarrow *$$

a-

$$p(\lambda) = \lambda^3 - 2\lambda^2 + \lambda + 5$$

$$p(0) = (0)^3 - 2(0)^2 + 0 + 5$$

$$p(0) = 5$$

put $p(0) = 5$ in *

$$\det(A) = 5$$

$(-1)^3 \rightarrow$ Power 3 as polyno has 3 as higher degree

$$\det(A) = -5 \text{ Ans}$$

b-

$$p(\lambda) = \lambda^4 - \lambda^3 + 7$$

$$p(0) = 0 - 0 + 7$$

$$p(0) = 7$$

put $p(0) = 7$ in *

$$\det(A) = 7$$

$$(-1)^4$$

$$\det(A) = 7 \text{ Ans}$$

Question: 25

Suppose the characteristic polynomial of some matrix A is found to be P
 $P(\lambda) = (\lambda - 1)(\lambda - 3)^2(\lambda - 4)^3$. In each part answer the question and explain your reasoning.

a) What the size of A ?

b) IS A invertible?

c) How many eigenspace does A have?

a-

Ans: 6×6 (Matrix Size)

Reasoning:-

Size of matrix is determined by degree of its characteristic polynomial counting powers.

$$(\lambda - 1)^1 (\lambda - 3)^2 (\lambda - 4)^3$$

Total Degree

$$1 + 2 + 3 = 6$$

Therefore A is 6×6 matrix.

b-

Ans: Yes, matrix A is invertible.

Reasoning:-

Matrix is invertible iff 0 is not an eigen value.

Eigenvalues are obtained from factors of characteristic polynomial.

$$\lambda - 1 = 0, \lambda - 3 = 0, \lambda - 4 = 0$$

$$\lambda = 1, \lambda = 3, \lambda = 4$$

Since 0 is not an eigenvalue, determinant is non-zero so A is invertible.

C-

Ans: 3 (Eigenspace)

Reason:

Each distinct eigenvalue has its own eigenspace.

The eigenvalues are

$$\lambda = 1, \lambda = 3, \lambda = 4$$

so there are 3 distinct eigenspaces.

Question : 27

Find 3×3 matrix A that has eigenvalues 1, -1, and 0 and for which

$$\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

are their corresponding eigenvectors

$$A = PDP^{-1}$$

$$P = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & -1 \\ 1 & 0 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

In Diagonal matrix the diagonal entries are always eigenvalues

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$$P^{-1} = ?$$

$$= \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ -1 & 1 & -1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \end{array} \right]$$

$$= \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 1 & 0 \\ 0 & -1 & -1 & -1 & 0 & 1 \end{array} \right] \begin{array}{l} R_2 + R_1 \\ R_3 - R_1 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & -1 & -1 & -1 & 0 & 1 \end{array} \right] \begin{array}{l} \\ R_2 \times \frac{1}{2} \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & -1 & -\frac{1}{2} & \frac{1}{2} & 1 \end{array} \right] \begin{array}{l} R_3 + R_1 \\ R_1 - R_2 \\ R_3 - R_2 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 & \frac{1}{2} & -\frac{1}{2} & -1 \end{array} \right] \begin{array}{l} \\ -R_3 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 & \frac{1}{2} & -\frac{1}{2} & -1 \end{array} \right] \begin{array}{l} \\ R_1 - R_3 \end{array}$$

$$P^{-1} = \left[\begin{array}{ccc} 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{1}{2} & -1 \end{array} \right]$$

PD = ?

$$= \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ -1 & 1 & -1 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$= \left[\begin{array}{ccc|ccc} 1+0+0 & 0-1+0 & 0+0+0 & & & \\ -1+0-0 & 0-1-0 & 0+0+0 & & & \\ 1+0+0 & 0-0+0 & 0+0+0 & & & \end{array} \right]$$

$$= \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & & & \\ -1 & -1 & 0 & & & \\ 1 & 0 & 0 & & & \end{array} \right]$$

$A = PDP^{-1}$

$(PD)P^{-1}$

$$A = \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 0 & 1 \\ -1 & -1 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & -1 \end{array} \right]$$

$$A = \left[\begin{array}{ccc|ccc} 0 - \frac{1}{2} + 0 & 0 - \frac{1}{2} - 0 & 1 - 0 - 0 & & & \\ 0 - \frac{1}{2} + 0 & 0 - \frac{1}{2} + 0 & -1 - 0 - 0 & & & \\ 0 + 0 + 0 & 0 + 0 - 0 & 1 + 0 - 0 & & & \end{array} \right]$$

$$A = \left[\begin{array}{ccc|ccc} -\frac{1}{2} & -\frac{1}{2} & 1 & & & \\ -\frac{1}{2} & -\frac{1}{2} & -1 & & & \\ 0 & 0 & 1 & & & \end{array} \right] \text{ Ans}$$